Received: 9 June 2025 | Accepted: 26 October 2025

DOI: 10.1111/mice.70144

INDUSTRIAL APPLICATION

COMPUTER-AIDED CIVIL AND INFRASTRUCTURE ENGINEERING WILEY

Enhanced precision in axle configuration inference for bridge weigh-in-motion systems using computer vision and deep learning

Domen Šoberl[1] | **Jan Kalin**[2] | **Andrej Anžlin**[2] | **Maja Kreslin**[2] | **Klen Čopič Pucihar**[1,3] | **Matjaž Kljun**[1,3] | **Doron Hekič**[2] | **Aleš Žnidarič**[2]

[1]Faculty of Mathematics, Natural Sciences and Information Technologies, University of Primorska, Koper, Slovenia

[2]Slovenian National Building and Civil Engineering Institute, Ljubljana, Slovenia

[3]Department of Information Science, Stellenbosch University, Stellenbosch, South Africa

Correspondence
Domen Šoberl, Faculty of Mathematics, Natural Sciences and Information Technologies, University of Primorska, Glagoljaška 8, 6000 Koper, Slovenia.
Email: domen.soberl@famnit.upr.si

Funding information
The Slovenian Research and Innovation Agency, Grant/Award Numbers: P5-0433, IO-0035, J5-50155, J7-50096; Research Programs CogniCom, Grant/Award Number: 0013103; University of Primorska and the Building Structures and Materials, Grant/Award Number: P2-0273; Slovenian National Building and Civil Engineering Institute

Abstract

Heavy goods vehicles (HGVs) have a significant impact on road and bridge infrastructure, with overloaded vehicles accelerating structural deterioration and increasing safety risks. Bridge weigh-in-motion (B-WIM) systems estimate gross vehicle weight (GVW) using strain measurements, but inaccuracies in axle configuration recognition can reduce reliability. This study presents a low-cost computer vision (CV) extension for existing B-WIM installations that verifies strain-inferred axle configurations using traffic camera images and flags GVW estimates as reliable or unreliable. Experiments on a data set of over 30,000 HGV records show that by combining convolutional neural networks with strain-based heuristics, GVW reliability can improve from 96.7% to 99.89%, effectively excluding nearly all erroneous measurements. The approach operates without interrupting ongoing B-WIM operations and can be applied retrospectively to historical data. Limitations include the inability to detect raised axles (RAs), which the method excludes as unreliable. This method provides a practical, high-precision enhancement for structural health monitoring of bridges.

1 | INTRODUCTION

Early applications of deep learning (DL) in civil engineering date back nearly a decade. Examples include the use of deep restricted Boltzmann machine (RBM) models to estimate concrete properties from mixture proportions (Rafiei et al., 2017) and the application of advanced machine learning models for construction cost estimation that incorporate physical and financial project factors, as well as economic variables and indexes (Rafiei & Adeli, 2018). More recently, DL techniques have been applied to vehicle load identification on bridges (Chang et al., 2025) and to improve the accuracy of vehicle weight estimation (Yan et al., 2025), which is crucial for detecting overloaded trucks and assessing bridge damage caused, in particular, by heavy goods vehicles (HGVs).

This is an open access article under the terms of the Creative Commons Attribution-NonCommercial-NoDerivs License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.
© 2025 The Author(s). *Computer-Aided Civil and Infrastructure Engineering* published by Wiley Periodicals LLC on behalf of Editor.

HGVs, as defined by the EU, are motor vehicles with at least four wheels with or without a trailer (i.e., road tractors, road tractors with semi-trailers, or lorries), with a permissible gross vehicle weight (GVW) of over 3.5 tonnes and are solely used for transporting goods (Slootmans, 2023). These vehicles can significantly accelerate the deterioration of roads and, to a lesser extent, bridges, particularly when fully loaded. Their size and weight exert significant stress on road surfaces and bridge structures, which reduces their lifespan, poses a safety risk to travelers, contributes to vehicle wear and tear, and reduces driving comfort (Meiring & Myburgh, 2015). Overloaded HGVs are especially problematic, compromising vehicle stability and safety, increasing the likelihood of road accidents (Shah et al., 2016). Furthermore, underpowered, overloaded vehicles cause longer overtaking times, lower uphill speeds, and higher fuel consumption (Kirushnath & Kabaso, 2018), while they also significantly accelerate damage to road infrastructure (Pais et al., 2013; Rys et al., 2016).

An established method for collecting HGV loading information and enforcing weight regulations is the use of weigh-in-motion (WIM) systems (Jacob & Feypell-de La Beaumelle, 2010; Sujon & Dai, 2021). These systems can record GVWs, axle loads, axle distances, and other important information about HGVs in a particular road section. Although WIM measurements are less accurate than traditional static weighing on dedicated platforms under controlled conditions, they are significantly faster than other weighing methods and do not require stopping the traffic. WIM is the only technology that captures load information for the entire traffic flow on a specific road section. Bridge WIM (B-WIM) (Moses, 1979) systems instrument existing bridges or culverts as scales to weigh the crossing vehicles. They do not require road closures during installation or maintenance and are more durable because no system components are in direct contact with the vehicles.

The weighing challenge of a traditional B-WIM system is to minimize the difference between the measured strains and the strains calculated using the system of equations:

$$s(t_j) = \sum_{i=1}^{N} A_i I\left[v_i\left(t_j - t_i\right)\right]; \quad j = 1 \ldots M \tag{1}$$

where $s(t_j)$ are the calculated strains at a time t_j, caused by a vehicle, A_i are the unknown axle weights of this vehicle, $I(x)$ represents the known influence line, v_i denote the axle velocities, t_i are the times at which individual axles arrive, N refers to the total number of axles, and M indicates the total number of strain measurements (Moses, 1979). An influence line describes the variation of a function, such as shear, bending moment, or strain, under the moving unit load. Figure 1 illustrates how the bending moment, or the proportional strain of a bridge at time t_j, is calculated using the known influence line and N axle weights that are not known in advance, in this case, from A_1 to $A5$ (Žnidarič et al., 2018). The influence line must be derived from measurements on the particular bridge to provide accurate results.

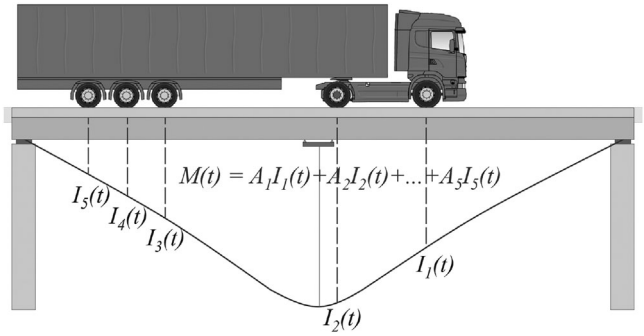


FIGURE 1 Calculation of theoretical bridge response with Equation (1), from Žnidarič et al. (2018).

Accurate axle recognition is essential for precise vehicle weighing. Initially, axles were detected using sensors either permanently embedded in the road's wearing course or temporarily mounted on its surface. The latter method was mainly used for mobile installations but had a short lifespan due to direct wheel impacts. Over the past two decades, surface-mounted axle detectors have largely been replaced by the free-of-axle-detector (FAD) approach, which derives axle positions and spacings from additional strain sensors installed on the underside of the bridge. This method offers a significant advantage over other WIM systems in eliminating the need for traffic closures during installation and maintenance. However, it provides less reliable axle information, as the data are acquired indirectly by sensors located on the underside of the superstructure, rather than in direct contact with the wheels.

Due to load transfer through the bridge superstructure, the signal peaks representing individual axles are smeared and often cannot be accurately located. As a result, axles may be misidentified, with some being missed and others incorrectly added. This problem has been mitigated using complex signal-processing algorithms (Žnidarič et al., 2018). As bridge length and traffic density increase, the system of equations for calculating axle loads gradually becomes ill-conditioned. One way to address this issue is by grouping closely spaced double or triple axle configurations to reduce the number of equations. The axle configuration of the truck in Figure 1 consists of three groups of axles: two groups with one axle each and one group with three axles, which can be denoted as a 113 configuration. If the axle configuration is correctly inferred from the strain data, the peaks in the strain signal can be used to estimate the GVW of the passing HGV, since the weight associated with a peak is distributed among the axles in the same group.

14678667, 2025, 30, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/mice.70144 by University Of Primorska Up Fvz, Wiley Online Library on [01/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Currently, there is no reliable estimate of the accuracy of the above method for inferring axle configurations from strain data in a real-world environment. There is also no method for estimating the reliability of individual GVW estimations from strain data. Such reliability assessment could help eliminate potentially unreliable outputs and improve the overall precision of the calculated GVWs, resulting in higher output credibility for B-WIM. In this paper, we propose a low-cost solution that does not require interrupting an operational B-WIM system. This solution double-checks the strain-inferred axle configurations using a traffic camera installed at the B-WIM site and flags each GVW as either reliable or unreliable. We show that the credibility of GVW estimation can be improved significantly by adding the proposed solution to an existing B-WIM site. We conducted experiments on a data set collected at a B-WIM site on a highway near Ljubljana, Slovenia, in 2014 and 2015, which contains over 30,000 HGV records.

The assumptions of this paper are as follows:

1. The proposed solution should be low-cost and involve little or no interruption of the existing B-WIM installation. It should be applicable retrospectively to data collected in the past, provided that a traffic camera was used at the site in question.
2. The accuracy of the GVW estimate depends on the correctness of axle configuration inference. We assume that correct axle group inference yields a reliable GVW estimate.
3. We are interested in estimating the weight of heavier vehicles. Discarding the GVWs of lighter or unloaded trucks is not considered a significant drawback.

The contributions of this paper are as follows:

(1) the assessment of the accuracy of existing B-WIM axle recognition heuristics;
(2) adaptation and assessment of existing DL architectures for recognizing axle configurations from low-resolution traffic camera photos;
(3) integration of the proposed DL method into an existing B-WIM installation to improve the credibility of GVW estimates.

The remainder of the paper is organized as follows: Section 2 provides an overview of related work in the field of WIM axle recognition. Section 3 defines the problem and formulates the research questions. The main contribution is presented in Section 4, where the proposed methods are described. The application of these methods to a specific data set is discussed in Section 5, and the results are presented in Section 6. Finally, key findings and limitations are discussed in Section 7, and the paper concludes in Section 8.

2 | RELATED WORK

B-WIM systems calculate the axle loads and GVW of HGVs crossing the bridge based on bridge deformation, which is typically measured with strain transducers. Except in cases where this is not possible (such as excessively long spans or very high superstructures), the strain transducers also provide information used to detect axles (Moses, 1979; Žnidarič & Kalin, 2020). FAD systems are inherently more robust, but the axle information they provide is less accurate than that from traditional wheel-triggered axle detectors, particularly at longer spans and with higher superstructures.

Recent advances in computer vision (CV) have enabled the development of alternative methods for determining vehicle parameters using traffic cameras. CV-based WIM can be broadly categorized into two groups: (1) non-contact WIM (NC-WIM) or visual WIM (V-WIM), which uses only a camera, and (2) hybrid systems that combine CV with strain data to improve weight estimation accuracy.

NC-WIM methods aim to determine all necessary weighing parameters by applying CV techniques to tire images captured by a roadside camera. The tire contact area is estimated by measuring the tire contact length along the road and the vertical tire deflection. Optical character recognition (OCR) is also used to obtain tire manufacturer information. The weight is then calculated as the sum of the individual tire–road contact forces (Feng et al., 2020). Feng and Leung (2021) initially assumed simple shapes for contact patches, such as rectangles or ovals. Kong et al. (2022) improved this by developing theoretical equations for tire contact forces and simulating the deformation process of different tire types in 3D. Zhang et al. (2023) proposed a tire deformation detection method based on CV and DL techniques. They created a data set of tire images from different types of vehicles and proposed a tire deformation calculation algorithm based on subpixel-level edge detection, keypoint positioning, and scale factor determination. They recently evaluated their approach in a real-world toll station setting with 894 HGVs (Zhang, OBrien, et al., 2024).

Zhang, Zhu, et al. (2024) argued that existing CV methods for extracting tire deformations are too computationally intensive and time-consuming. They proposed using the YOLOv5 model to segment tires and extract key regions of interest (ROIs), such as the contact line, wheel rim, and text on the tire sidewall. This method achieved up to 17 times faster extraction of the tire–road contact line compared to the method proposed by Feng and Leung

14678667, 2025, 30, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/mice.70144 by University Of Primorska Up Fvz, Wiley Online Library on [01/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

(2021). Almutairi et al. (2022) reported achieving above 97% precision in recognizing individual tires using various versions of YOLO. Gao et al. (2024) used a thermal imaging camera to extract tire deformation under different lighting and weather conditions. They combined a thermal image–based edge detection algorithm to detect tire deformation and a YOLOv5-based sidewall marker detection to obtain tire information. Feng et al. (2024) attempted a similar thermal camera approach and also applied explainable machine learning methods. They found that the most informative feature for training a vehicle weight regression model is the ratio of tire–ground contact length.

Recognizing individual tires, even with high accuracy, does not guarantee correct axle configuration inference, which depends on a carefully calibrated transformation function that converts pixel measurements into spatial coordinates (Xu et al., 2023; Yang et al., 2024). Instead, axle types can be inferred directly through DL. The truck's taxonomy, recognized from a photo, could serve as a lookup reference in a database of trucks. Almutairi et al. (2022) reported 96% test accuracy when classifying 13 classes of trucks. However, this does not detect the actual axles. Wang et al. (2024) used an unmanned aerial vehicle (UAV) with a high-resolution camera to detect vehicle axle type, reporting 97.1% average precision. However, using drones may be feasible for specific use cases and is likely too difficult and costly for use at a commercial B-WIM site.

We propose a low-cost DL approach that extends existing B-WIM sites without interrupting ongoing operations and can also be applied to legacy B-WIM data. The model infers axle configurations directly from images captured by traffic monitoring cameras. Instead of relying solely on the classifier's output, we integrate it with existing B-WIM heuristics to identify and filter out incorrectly estimated GVWs.

3 | PROBLEM DEFINITION

B-WIM systems currently rely on manually fine-tuned heuristics to infer axle configurations from strain signals. Figure 2 shows an actual event at a B-WIM site, where sensors captured a passing HGV. The sensors installed beneath the bridge measured displacements along the influence lines. The signals are processed through multiple filters, each fine-tuned for the specific site, which are depicted as multiple lines on the plot in the figure. A set of heuristic rules, determined by experts, is used to infer the positions of the truck's axles, which are denoted by vertical lines on the plot. Finally, based on predetermined thresholds, axles that are close together are grouped. The axle

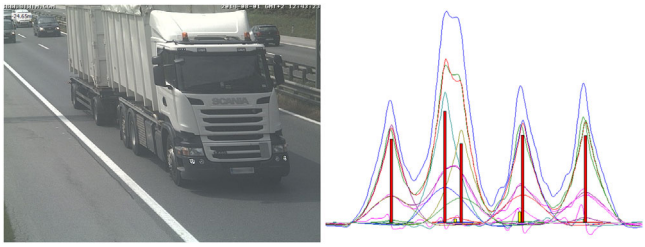


FIGURE 2 Strain signals (right) measured when an HGV (left) crossed the B-WIM measuring site. The vertical bars denote heuristically determined weights of individual axles.

configuration seen in Figure 2 is depicted as 1211, which means that the second and third axles are grouped, while all other axles are single. The plots in the figure also show that the second wave of signals is higher than the rest, because more weight was applied by this group of axles on the sensors. Information that the second wave belongs to a group of two axles, rather than a single axle, is therefore crucial to correctly interpret the signals.

If the B-WIM system determines that the vehicle is not an HGV (based on the distance between the axles), the event is discarded and the vehicle's weight is not estimated. In this way, all cars and other smaller vehicles are ignored. The system also ignores the event if the presence of multiple vehicles is recognized from the strain data.

A low-resolution traffic monitoring camera is typically installed at a B-WIM site to capture a photo of the HGV, which is then processed by the system. The position and angle of the camera, as well as the quality of the captured photos, are not necessarily ideal for automated recognition of the vehicle's axles. However, it should still be possible to determine the vehicle's axle configuration by eye. As seen in Figure 2, the visual conditions at the site used in this research are far from perfect; therefore, the results presented in this paper should be achievable at other B-WIM sites with similar or better conditions.

We propose a CV-based axle group detection method that classifies traffic camera images into predefined classes, each representing an axle configuration such as 113, 1211, and so on. We use machine learning to avoid manual fine-tuning of parameters for specific sites, thereby eliminating tedious and error-prone expert intervention. We run the proposed method in parallel with the existing B-WIM system, as shown in Figure 3. Both systems independently infer the axle configuration, and if they agree on the output class, the estimated GVW is flagged as reliable. The goal is to reduce the number of incorrectly estimated GVWs by discarding those flagged as unreliable, which, in machine learning terminology, corresponds to the performance metric known as *precision*, where correctly estimated reliable instances are considered

14678667, 2025, 30, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/mice.70144 by University Of Primorska Up Fvz, Wiley Online Library on [01/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

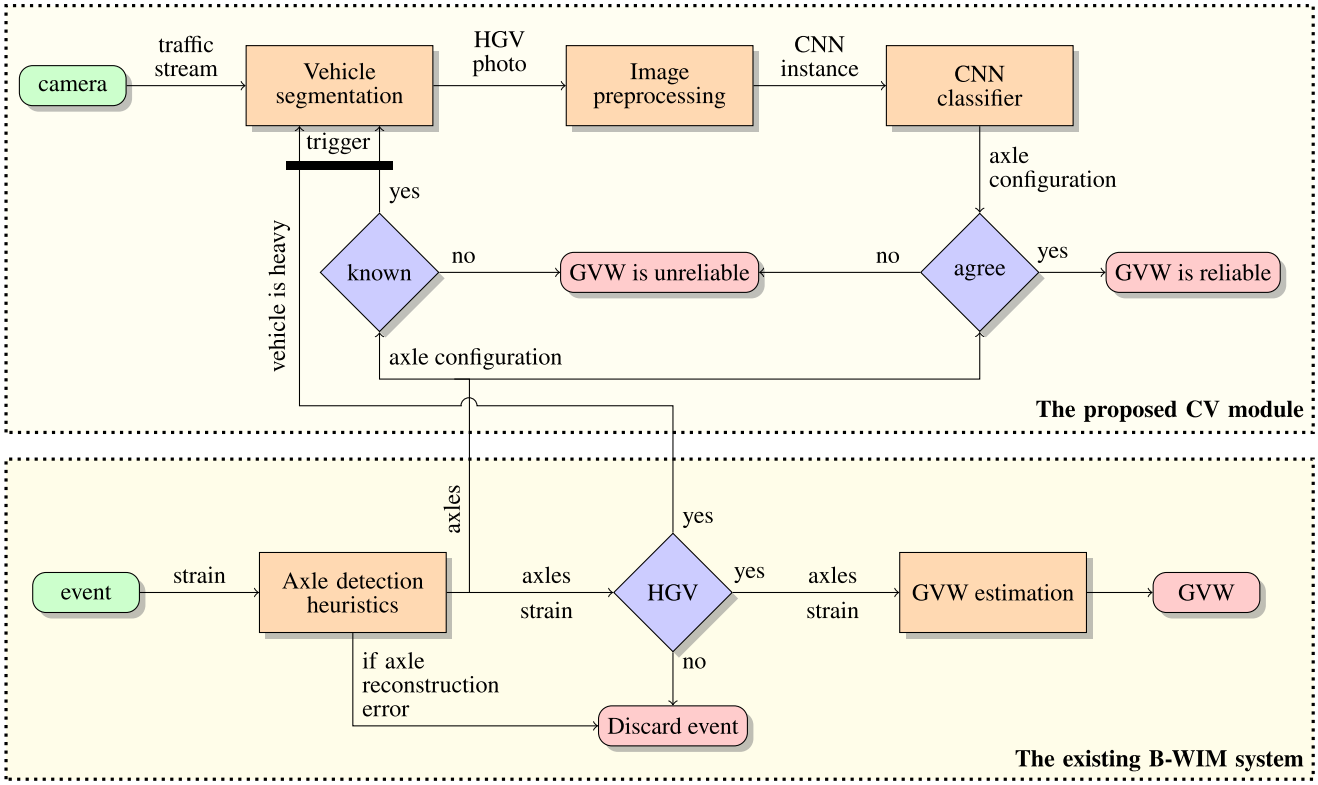


FIGURE 3 The proposed improved B-WIM axle configuration recognition method.

true positives and incorrectly estimated reliable instances are considered *false positives*. This differs from using a multimodal approach, where both sources are input to the same model. Keeping the models separate lowers the possibility of correlated noise and allows for more conservative testing by cross-validating both outputs.

We identify two significant limitations to the proposed approach in advance. First, the traffic monitoring cameras cannot capture useful images at night or in poor weather conditions, such as fog or heavy rain. Vehicles passing the site under these conditions will not be considered for GVW estimation by the proposed method, as they will automatically be flagged as unreliable. However, the significant drop in HGV traffic during nighttime mitigates this problem.

The second limitation concerns raised axles (RAs). When an HGV is lightly loaded or empty, drivers lift certain axles to reduce fuel consumption and tire wear. While strain sensors will not register an RA, the camera will detect it. This discrepancy between the two axle configurations would flag the result as unreliable, even if the strain-based computations are correct. However, since we are primarily concerned with measuring the weight of heavily loaded vehicles, the exclusion of HGVs with RAs does not necessarily represent a significant drawback.

We formulate the following research questions:

1. With what accuracy can axle configurations of HGVs be recognized using DL methods?
2. How much data are needed to train a DL model for axle configuration recognition?
3. How much can the precision of strain-based B-WIM axle configuration recognition heuristics be improved by incorporating a traffic camera and DL methods?

4 | METHODS

This section describes the methods used in the proposed extension to the existing B-WIM system, as shown in Figure 3. When the B-WIM system infers an axle configuration from the strain data and the heuristic rules determine that the strain event is caused by an HGV, the CV module is activated. If the axle configuration falls within the classification scope of the CV module (some very rare axle configurations may not be detectable through our CV method), the photo taken at the moment of the strain event is processed through a three-stage pipeline: vehicle segmentation, image preprocessing, and axle group classification, which is performed using a pretrained convolutional neural network (CNN). If the CNN recognizes the same axle configuration as B-WIM, the instance is flagged as reliable; otherwise, it is flagged as unreliable.

14678667, 2025, 30, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/mice.70144 by University Of Primorska Up Fvz, Wiley Online Library on [01/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

FIGURE 4 The process of generating training instances from the raw camera feed.

4.1 | Vehicle segmentation

The vehicle segmentation stage involves identifying the type of vehicle and computing its bounding boxes. This can be accomplished using the YOLO (You Only Look Once) object detection and segmentation model (Redmon et al., 2016). We used the pretrained YOLOv8 model, which can detect 80 different object classes, including various types of motorized road vehicles such as cars, trucks, buses, and motorcycles. In our case, no additional training was necessary, as the pretrained weights performed well on our data. We considered only segments tagged as "truck" or "bus" and ignored the rest.

Although the B-WIM system recognizes and discards events with multiple vehicles present, YOLO will detect other vehicles outside the range of the strain sensors, typically HGVs far behind the measured one or those on the opposite side of the highway, driving in the opposite direction. This issue can be addressed easily, since the measured vehicle is closest to the camera and therefore occupies the largest area in the photo. In cases with multiple HGVs in the photo, the segment covering the largest area is selected. An example is shown in Figure 4, where the camera captured two HGVs driving one after another. The first was correctly identified as the one being measured.

There are rare cases where YOLO fails to correctly detect or segment the measured HGV. Most often, YOLO segments only the tractor instead of the entire tractor-and-trailer composition, effectively splitting the truck in half. As a result, only the front part of the HGV is processed for axle detection, leading to a class with fewer axles. In our experiments, about 1% of images were incorrectly classified due to YOLO errors. These cases are flagged as unreliable because of a mismatch between the CV and strain-based inference of axle configurations.

During nighttime or under very bad weather conditions (such as thick fog), YOLO fails to recognize and segment the HGV entirely. In such cases, the event is discarded and not considered for GVW estimation.

4.2 | Image preprocessing

The traffic camera used with our B-WIM system captured images at a 640 × 480 pixel resolution in RGB (24-bit) color format. After cropping the segmented vehicle, the images used for classification are small, rarely exceeding 200–300 pixels in width or height. However, many popular CNN models require RGB images with a resolution of 224 × 224 pixels as input, which is close to our original image resolution. This usually requires only slight resizing of the cropped part to fit the required 224 × 224 format.

To tackle overfitting during CNN training, image preprocessing may include slight distortions, such as random scaling and horizontal or vertical shifting. We apply these modifications separately to each batch of training samples, so the samples differ slightly in each training epoch. However, we limited these modifications to avoid losing too much visual information through scaling or shifting an axle beyond the visible edge of the image. The images were scaled by a random factor between 0.9 and 1.0, then randomly shifted horizontally and vertically within the borders of the original 224 × 224 pixel area. We did not use image rotation, as the traffic camera is mounted at a fixed orientation. Rotation would train the CNN for viewing angles that are irrelevant for inference.

An example of image preprocessing is shown in Figure 4. First, YOLO is used to detect and segment an HGV (a). If multiple HGVs are recognized (the two rectangles in the second image), the one with the largest is selected. The image is then cropped and scaled to match the 224 × 224 format (b). Finally, random scaling and translations are applied to introduce variation into the training process (c).

4.3 | CNN classifiers

We trained and evaluated five popular classification models: VGG16 and VGG19 (Simonyan & Zisserman, 2015), DenseNet (Huang et al., 2017), MobileNet (Howard et al., 2019), and ResNet (He et al., 2016). The selection was based

14678667, 2025, 30, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/mice.70144 by University Of Primorska Up Fvz, Wiley Online Library on [01/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

on two factors: (1) each architecture accepts input images of 224×224 pixels with three color channels (RGB), a widely adopted format compatible with popular image data sets, and (2) a pretrained implementation is available in the TensorFlow/Keras library (Abadi et al., 2016). We therefore focus on established, widely adopted industry-standard CNN architectures that are publicly accessible and available as pretrained models.

The chosen classification models are based on CNNs. A CNN is a DL architecture specifically designed to process data with a grid-like structure, such as images. It consists of multiple convolutional layers that apply learnable filters to extract local features, pooling layers that reduce spatial dimensionality while retaining key information, and fully connected layers that learn to translate the extracted image features into output classes. The popular choice of a 224×224 input format is a reasonable trade-off between the complexity of the CNN architecture and sufficiently distinctive visual information, while the input dimension 224 can be divided by 2 several times within the architecture.

We accommodate the size of the final layers to match the number of distinct axle configurations our system can recognize. After experimentation, we found that adding a dense layer with 512 neurons after the convolutional layers, followed by a drop out layer with a rate of 0.5 to improve generalization, and an output layer with the number of distinct classes, works best for this domain. This is a notable reduction from their original sizes, therefore, the number of trainable parameters listed in this paper for each architecture differs from the numbers in the original publications.

All five CNN architectures are available as pretrained models, capable of recognizing 1000 different object categories, such as "balloon," "castle," "necklace," from the ImageNet (Deng et al., 2009) database. These categories are not relevant to our domain, so additional learning (transfer learning) is needed to adapt the network to our axles domain. However, the pretrained weights in the initial convolutional stages, which learn low-level image features, might accelerate the training process. We compare the speeds of the five models when training from scratch versus using transfer learning.

4.4 | Training the models

The models are trained on a large set of manually labeled photos of HGVs. Each photo was examined by an expert who determined its true axle configuration based on their knowledge of B-WIM heuristics, primarily the thresholds for axle groups. Very rare axle configurations may be ignored, meaning these will automatically be flagged as unreliable once the system is deployed. In our data set, we observed a few unusual axle configurations that appeared only a few times over several years. Ignoring these does not pose a significant disadvantage in practice.

We considered introducing an additional class called *other*, to which unknown axle configurations would be assigned. However, our experiments showed an overall decrease in classification accuracy (CA) when using the *other* class, indicating that the neural network failed to learn the abstract meaning of this category. Many known instances that were previously classified correctly were predicted as *other* when this class was introduced. Moreover, even if a predictor correctly learns to use such a category, it cannot be compared with the B-WIM's prediction, so the GVW estimate would be flagged as unreliable anyway.

After acquiring the labeled data set, classes with sufficient representation for training, as well as their order, are determined; for example, $C_0 = 113, C_1 = 1211, C_2 = 122$. One-hot encoding is used to encode the ground truth, assigning the value 1.0 to the true class C_k, and 0 to all other classes. The class label C_k is thus represented as $[C_0 = 0, C_1 = 0, \dots, C_k = 1, \dots, C_n = 0]$ within the training set. During inference, the output vector represents a probability distribution over the classes, and the instance is classified into the class with the highest probability.

To train the neural network, we used the categorical cross-entropy loss function, defined as:

$$L(y, \hat{y}) = -\sum_{i=1}^{n} y_i \log(\hat{y}_i) \tag{2}$$

where y is the one-hot encoded ground truth and $\hat{y}$ is the output class probability distribution. To minimize the loss during training, we used the Adam optimization method (Kingma & Ba, 2015) with the learning rate set to $\alpha = 0.0001$, which is a typical choice for this type of optimization problem and performed well with our data set.

4.5 | Evaluation

To evaluate the improvement in B-WIM precision, we use the evaluation metric shown in Figure 5. The baseline performance is the *CA* of the strain-based B-WIM heuristics. With the ground truth available, we can determine the percentage of correctly inferred axle configurations, denoted as C in the figure, while incorrect inferences are denoted as I.

Introducing the proposed CV-based extension, referred to here as the *discriminator*, the strain-based inferences are flagged as either reliable (positive) or unreliable

14678667, 2025, 30, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/mice.70144 by University Of Primorska Up Fvz, Wiley Online Library on [01/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

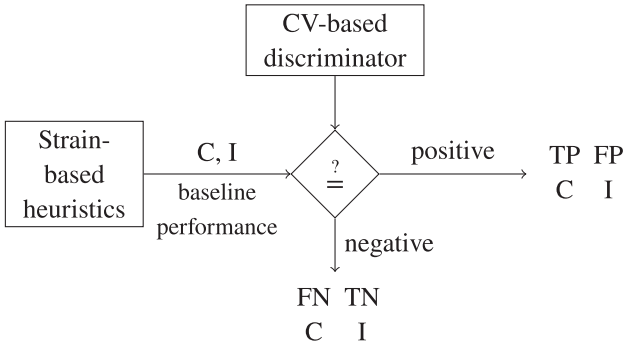


FIGURE 5 Measuring the performance of the proposed B-WIM system.

(negative). This results in four types of events that align with the machine learning concepts of *true positive* (TP), *false positive* (FP), *false negative* (FN), and *true negative* (TN) predictions. For example, if the strain-based system correctly infers the axle configuration and the discriminator agrees with its output, the event is considered a *true positive*. If the discriminator disagrees with that correct output, it is a *false negative* event, and so on. We measure *precision* and *recall*, defined as:

$$\text{precision} = \frac{TP}{TP + FP} \tag{3}$$

$$\text{recall} = \frac{TP}{TP + FN} \tag{4}$$

High precision means that the number of incorrect axle configurations within the *reliable* set is low, allowing us to have a high level of confidence in the correctness of the computed GVWs. High recall means that our process of elimination did not exclude many correct instances. Note that by removing the discriminator, we have FN = 0 and TN = 0, so the CA of the baseline strain-based heuristic equals its precision, while its recall is 100%. Therefore, we aim to improve the precision of the baseline system while trying to keep the reduction in recall minimal.

5 | APPLICATION

The proposed methods were applied to a data set collected at a highway near Ljubljana, Slovenia, between 2014 and 2015. The B-WIM system installed under an 8-m-long slab bridge recorded its responses during the HGVs crossing the bridge, while a traffic camera mounted on the bridge captured still images of the traffic whenever an HGV triggered the predefined strain levels.

TABLE 1 Distribution of the available data samples by classes.

	Labeled samples		Validation set	
Class	**Number**	**Percentage**	**Number**	**Percentage**
113	11,921	38.98%	524	47.51%
1211	4550	14.88%	25	2.27%
122	3705	12.12%	56	5.08%
11	2764	9.04%	340	30.83%
22	2440	7.98%	7	0.63%
111	1611	5.27%	9	0.82%
112	1516	4.96%	105	9.52%
1112	669	2.19%	4	0.36%
12	460	1.50%	24	2.18%
1111	331	1.08%	3	0.27%
123	267	0.87%	4	0.36%
1212	219	0.72%	1	0.09%
1222	128	0.42%	1	0.09%
Unused	309	1.01%	5	0.45%

5.1 | Data labeling

The experts manually examined 30,890 photos of HGVs and determined their true axle configurations. They identified 44 different classes, but most were heavily underrepresented—11 contained only one vehicle and 23 contained fewer than 10. We included only classes with at least 100 vehicles to ensure some sample variety, finally resulting in 13 classes and 30,581 training instances. Their distribution is shown in Table 1.

The data set is heavily imbalanced, with most vehicles belonging to class 113—the 2-axle tractor with a 3-axle semi-trailer. However, this issue is addressed during training by upsampling the classes using the image preprocessing method shown in Figure 4.

Note that the distribution of the labeled data does not necessarily represent the real-world distribution of vehicle types, as the labeling was done arbitrarily. A more accurate approximation of the actual distribution can be obtained by analyzing data collected over a specific period, which we have done when selecting the validation set.

5.2 | Validation set

As the validation set, the day with the highest count of labeled instances was chosen, which was March 10, 2014. There were 1691 events recorded on that date, of which YOLO recognized an HGV in 1256 cases, failing mostly during nighttime. The distribution of events on that day by hour is shown in Figure 6.

There were 15 cases where YOLO recognized an HGV but failed to segment it correctly. The error was either

14678667, 2025, 30, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/mice.70144 by University Of Primorska Up Fvz, Wiley Online Library on [01/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

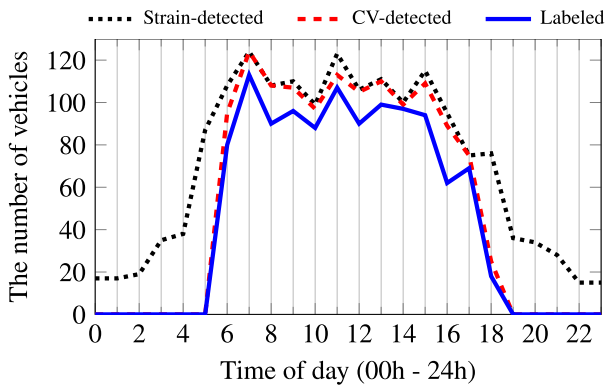


FIGURE 6 The hourly distribution of HGVs on the validation day: Detected by the existing B-WIM strain-based system (dotted), segmented by YOLO (dashed), and labeled by our experts (solid).

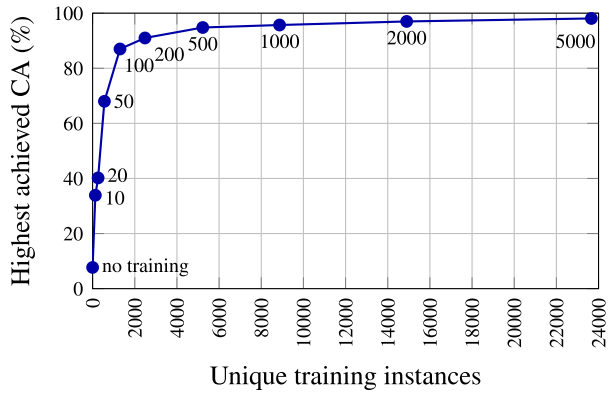


FIGURE 7 The highest achieved classification accuracy (CA) for different sizes of the training set.

cutting off the tractor from the trailer or missing the frontal part of the vehicle in the photo, making correct segmentation impossible. The latter case was due either to incorrect camera timing or the vehicle driving above the presumed speed. There were 133 cases where YOLO correctly segmented the HGV, but our experts could not determine the ground truth, either because of poor visibility or partial axle occlusions. Most of these cases occurred at night, when traffic headlights allowed YOLO to detect and segment an HGV, but the axles were not sufficiently illuminated for the expert to identify the correct class. Five vehicles had rare axle configurations not recognizable by our trained models: 2222, 121, 21, 3, and 114. We were left with 1103 labeled validation instances, whose distribution by class is shown in Table 1.

6 | RESULTS

We conduct four types of experiments:

1. *Sample efficiency* to assess the minimum number of instances needed to successfully train the models.
2. *Tenfold cross-validation* to assess the CA of the trained CNN models independently of the B-WIM framework.
3. *A separate validation* to assess the improvement in performance of the proposed system over an actual 24-h operational period.
4. *An ablation study* to evaluate the impact of sample augmentation on the final B-WIM performance and the impact of using pretrained weights on the speed of training.

All experiments were conducted on an NVIDIA RTX 4080 GPU with 16 GB of VRAM, using TensorFlow machine learning library version 2.17.0.

6.1 | Sample efficiency

Sample efficiency was evaluated by gradually increasing the training set size and recording the highest CA achieved by any of the five CNN architectures. We selected 10 subsets of different sizes as they became available during the lengthy manual data labeling process, using the following approach. For a predefined maximum class size S, we randomly selected S instances from each class. If a class contained fewer than S instances, all available instances from that class were used. All remaining instances were used for testing. No upsampling was performed for classes with fewer than S samples, so sample efficiency was measured only on unique samples.

We sampled 10 training subsets using $S = 10, 20, 50, 100, 200, 500, 1000, 2000$, and 5000. The CA for each training set is shown in Figure 7. The x-axis represents the number of samples used. The numbers next to the dots indicate the values of S. About 2000 unique instances are sufficient to achieve over 90% CA, after which progress slows significantly.

6.2 | CNN performance

The stand-alone CNN performance was evaluated separately for all five CNN architectures using stratified 10-fold cross-validation. To avoid training a biased model, underrepresented classes were upsampled by duplicating existing samples until each class contained 5000 samples. The 5000 samples from the 113 class were selected randomly. Sample augmentation was performed automatically on each training batch of 32 samples by the image preprocessing method presented in Figure 4. Transfer learning was conducted on models pretrained on the ImageNet database. We did not apply the incremental unfreezing

14678667, 2025, 30, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/mice.70144 by University Of Primorska Up Fvz, Wiley Online Library on [01/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

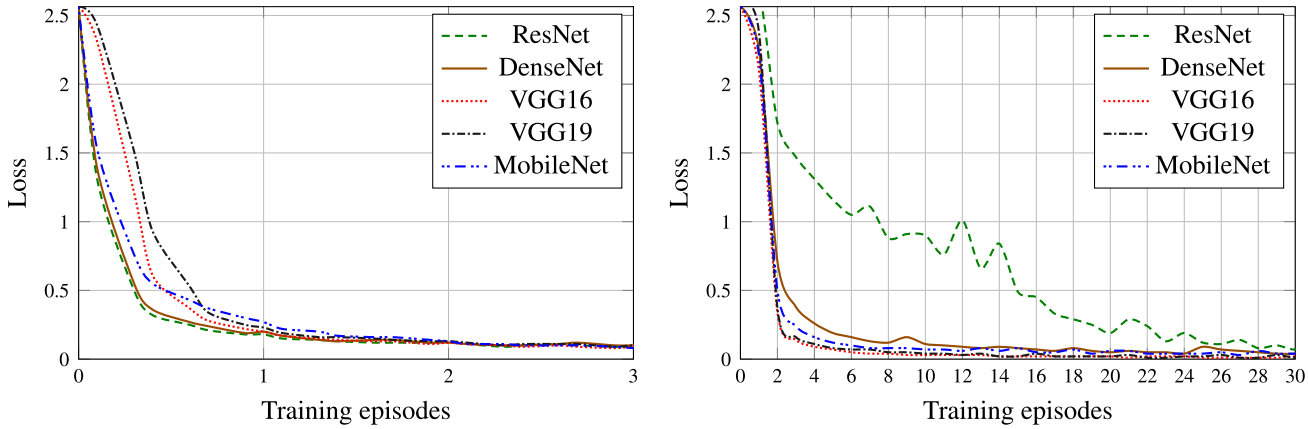


FIGURE 8 Training progress when conduction transfer learning (left) and when training the models from scratch (right).

TABLE 2 Classification accuracy (CA) and B-WIM performance of the five convolutional neural networks.

Base CNN architecture	Trainable parameters (millions)	Batch training time	Sample classification time	CA		B-WIM performance		
				10-fold cross-validation	24-hour validation	precision	recall (incl. RA)	recall (excl. RA)
VGG16	27.6	127 ms	45 ms	97.90% (1.91%)	98.01%	99.89%	87.45%	98.63%
VGG19	32.9	148 ms	48 ms	98.08% (1.20%)	97.55%	99.68%	86.61%	97.68%
ResNet101V2	93.9	108 ms	65 ms	96.74% (2.58%)	98.10%	99.79%	87.45%	98.52%
DenseNet121	32.7	94 ms	72 ms	97.47% (1.60%)	94.74%	99.56%	84.64%	95.26%
MobileNetV3	15.4	44 ms	57 ms	92.99% (5.18%)	92.29%	99.66%	83.33%	93.39%

technique, but instead unfroze all layers at the beginning of training. We did not find it necessary to perform any such fine-tuning, as training converged in just a few episodes, as shown by the left plot in Figure 8.

CA achieved by each of the five models is shown in Table 2. The *10-fold cross-validation* column shows the average CA over the 10 folds, with the standard deviation given in parentheses. The *24-h validation* column shows the results achieved on the validation set that was excluded from training. Note that the main difference between the two evaluations is the class balance. In both cases, the models were trained on a balanced (upsampled) data set, but the validation data set is unbalanced and reflects the actual class distribution on site. Both VGG-based models showed the most reliable performance and resilience to class distribution.

Table 3 shows the cumulative confusion matrix for the validation set, where the classifications from all five neural networks are summed. Thus, the matrix displays $5 \cdot 1103 = 5515$ classifications. This allows us to examine the most common errors made by convolutional classification in general. There are three types of errors: missing an axle, adding a nonexistent axle, or joining axles that should be recognized as separate.

Lastly, we may consider the size of the models in relation to their accuracy. We observe that the number of training parameters does not directly correlate with CA. Larger models with more parameters do not necessarily outperform smaller ones, indicating that performance depends more on architectural design choices than on model size. In this regard, the choice of VGG16 among the five tested seems most reasonable, as its performance is similar to VGG19, while its smaller size allows for slightly faster training and inference. Moreover, our experiments in either enlarging or shrinking the VGG-based structure did not yield better results than those presented.

There seems to be an interesting anomaly in Table 2, where MobileNet takes more time to infer a single instance than to train on a batch of 32 instances. This likely occurred due to poor GPU parallelism utilization when processing a single instance. Running the same experiment on a CPU resulted in training times about 30 times longer than inference.

14678667, 2025, 30, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/mice.70144 by University Of Primorska Up Fvz, Wiley Online Library on [01/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 3 Confusion matrix of the validation set, cumulative for all five convolutional neural networks. True values are in rows, predictions are in columns. The most noticeable errors are shaded.

	113	1211	122	11	22	111	112	1112	12	1111	123	1212	1222
113	2589	1	1	0	6	1	14	0	0	0	8	0	0
1211	0	118	7	0	0	0	0	0	0	0	0	0	0
122	2	0	263	2	7	0	6	0	0	0	0	0	0
11	0	0	1	1664	4	10	4	0	17	0	0	0	0
22	0	0	0	0	35	0	0	0	0	0	0	0	0
111	0	0	0	0	0	45	0	0	0	0	0	0	0
112	3	0	9	3	11	5	492	0	0	2	0	0	0
1112	1	0	1	0	1	0	0	17	0	0	0	0	0
12	0	0	0	4	10	0	0	0	106	0	0	0	0
1111	0	0	0	0	3	0	0	0	0	12	0	0	0
123	0	0	2	0	1	0	2	0	0	0	15	0	0
1212	0	1	0	0	0	0	0	0	0	0	0	4	0
1222	0	0	0	0	0	0	0	0	0	0	0	0	5

TABLE 4 B-WIM performance on the validation set using different CNN architectures (lower is better). 'Incorrect GVW' (bold values) is the key metric for evaluating improvements over the baseline method.

Method	Incorrect axles	Incorrect GVW	Lost all	Lost non-RA
VGG16	22	**1**	134	13
VGG19	27	**3**	143	22
ResNet101V2	21	**2**	134	14
DenseNet121	58	**4**	164	45
MobileNetV3	85	**3**	178	115
Baseline	35	**35**	0	0

6.3 | The improved B-WIM performance

Measuring the baseline CA of the existing B-WIM strain-based heuristics is straightforward. Of the 30,890 manually labeled instances, the existing B-WIM heuristic correctly inferred 29,550 instances, resulting in a 96.70% CA. In the validation set, 1068 out of 1103 labeled instances were correctly inferred, which is a 96.83% CA. This means that the GVW of 35 vehicles was certainly incorrectly estimated on the validation day.

The *B-WIM performance* columns in Table 2 show how much different CNN architectures improve the baseline B-WIM precision of 96.70%, as well as how much the recall decreases from the baseline of 100%, depending on whether vehicles with RAs are considered. Table 4 presents the B-WIM performance on the validation set as specific case counts in terms of losses, which means lower number is better. The *incorrect axles* column indicates how many of the 1103 labeled cases were incorrectly identified by the CNN alone. The *incorrect GVW* value, which is the focus of this research, shows how many of the "reliable" GVWs were incorrectly inferred. The *lost all* columns indicate how many GVW cases would have been correctly inferred but were flagged as "unreliable." The number of those cases without an RA is shown in the last column. These are potentially the heavier vehicles that may be of interest for weighing.

Statistical analysis of precision on the validation set is presented in Table 5. A paired bootstrap significance test with 5000 resampling iterations is used to compare improvements in B-WIM precision when using different CNN models. The table presents selected comparisons, as showing all possible pairs would be too exhaustive.

The results indicate that using the CV-extension with any of the five CNN models significantly outperforms the baseline B-WIM performance. There is no significant difference among the top three performing models: VGG16, VGG19, or ResNet. DenseNet and MobileNet, which showed notably lower CA during validation, also show a statistically significant drop in B-WIM precision ($p < .05$) when used instead of the top three. However, the evidence ($p = .0388$) is not very strong.

6.4 | Ablation study

We study the effect of the pretrained weights on the speed of training the models, and the effect of sample augmentation shown in Figure 4 on CA.

6.4.1 | Transfer learning

There is a notable difference in training speed between training models from scratch and performing transfer learning on models pretrained on the ImageNet database. However, both approaches achieve similar results after

14678667, 2025, 30, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/mice.70144 by University Of Primorska Up Fvz, Wiley Online Library on [01/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 5 Results of the paired bootstrap significance test with 5000 resampling iterations. The table reports the statistical significance in improving the B-WIM's precision using different CNN models. The *baseline* method uses no CV-based extension.

Compared methods		Mean difference	p	Significance ($p < .05$)	95% Confidence interval	
					Lower	Upper
VGG16	VGG19	0.2163	.1216	No	0.0000	0.5467
VGG16	ResNet101V2	0.1067	.5292	No	−0.0007	0.3247
VGG19	ResNet101V2	0.1096	.4228	No	−0.2146	0.5342
ResNet101V2	DenseNet121	0.2269	.0388	Weak	0.0026	0.5627
DenseNet121	MobileNetV3	0.3336	.0388	Weak	0.0030	0.7709
VGG16	Baseline	3.0662	<.001	Strong	2.0714	4.1525
VGG19	Baseline	2.8499	<.001	Strong	1.9039	3.8801
ResNet101V2	Baseline	2.9595	<.001	Strong	1.9688	4.0421
DenseNet121	Baseline	2.7326	<.001	Strong	1.7633	3.8017
MobileNetV3	Baseline	3.0662	<.001	Strong	2.0714	4.1525

TABLE 6 Change in classification accuracy (CA) on the validation set with and without sample augmentation, and its statistical significance.

Base CNN architecture	CA augmented	CA not augmented	χ^2	p
VGG16	98.01%	94.47%	30.72	<.001
VGG19	97.55%	93.74%	22.12	<.001
ResNet101V2	98.10%	92.75%	50.21	<.001
DenseNet121	94.74%	95.47%	0.98	.3222
MobileNetV3	92.29%	86.94%	114.38	<.001

a sufficient number of training epochs. Therefore, we compare only the decrease of loss values over training time.

Transfer learning converged by the third training episode in all five cases, as shown in the left plot in Figure 8. In contrast, the training progress when training the models from scratch is shown in the right plot in Figure 8. The training progress is noticeably slower but eventually achieves the same results as the pretrained approach. VGG-based architectures were the fastest, reaching results similar to their pretrained counterparts after about 10 episodes. DenseNet and MobileNet caught up with their pretrained counterparts in about 30 episodes. ResNet showed the worst performance but managed to catch up by episode 50. This can be attributed to its massive size.

6.4.2 | Sample augmentation

Table 6 shows the change in CA when upsampling underrepresented classes is done by duplicating existing images without applying the scaling and translation described in Section 4.2. McNemar's test was applied to assess whether the difference in CA between the two upsampling approaches was statistically significant.

(a) Poor visibility (b) Loaded cars

FIGURE 9 Difficult, but correctly classified by all models.

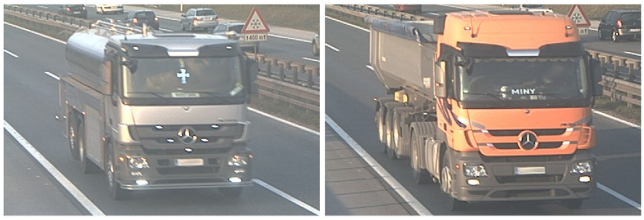
As shown in the table, CA decreased by a few percent when augmentation was not applied, except for DenseNet, which did not show any significant change ($p = .3222$). This provides strong evidence for the use of sample augmentation, which enables better knowledge generalization and does not have a detrimental effect on performance in the worst case.

6.5 | Case studies

We present several case studies of correct and incorrect image classifications. Figure 9 shows two challenging examples that were correctly classified by all five CNN models. Case (a) is barely recognizable to the naked eye due to foggy conditions. However, the information is present and discernible to the CNN. Case (b) shows an HGV carrying other vehicles, and the photo includes the tires of the loaded cars, which should not be considered part of the axle configuration. Numerous such cases demonstrate that the neural network learned to correctly focus on the HGV's wheels while ignoring those of the loaded cars.

Figure 10 shows two trucks with RAs. Strain signals do not register RAs, while the CNN includes them in the

14678667, 2025, 30, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/mice.70144 by University Of Primorska Up Fvz, Wiley Online Library on [01/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License



(a) Last axle raised (b) 3rd axle raised

FIGURE 10 Trucks with raised axles got flagged "unreliable."

FIGURE 11 Threshold error. In both cases, the last two axles should not be grouped by CNN.

inferred axle configuration. Both systems are correct in their own way but disagree on the axle configuration. Cases (a) and (b) in the figure are recognized by the CNN as 12 and 113, respectively, while strain-based heuristics infer 11 and 112. Such cases are therefore flagged as "unreliable" and excluded from GVW estimation. However, they cannot be regarded as erroneous.

There were many cases where CNN classification did not match the expert's label, but upon closer inspection, it becomes clear that the difference between the label and the inferred class lies in the threshold that defines an axle group. Typically, the CNN would group axles that appear close together, but according to experts, they did not meet the grouping threshold. This issue is largely due to the sharp camera angle. With the two cases showed in Figure 11, CNN grouped the last two axles, while they were labeled as separate.

There were also genuine CNN errors, which can be attributed to poor model performance. The model would either infer an axle where there is none (a "ghost axle") or miss an existing axle. Examples of both occurrences are shown in Figure 12. Typically, a shadow or part of the vehicle's body would be mistaken for an axle, or an axle would be missed due to poor visibility or partial occlusion. The latter case is largely due to the sharp camera angle.

Errors made by CV usually result in disagreement on the inferred axles and flagging the instance as "unreliable." This represents either a true or a false negative instance, so it does not negatively affect precision. Loss of precision occurs when both systems—the B-WIM's baseline heuristics and the proposed CV extension—make the same

(a) 'Ghost axle' Recognized as 111 (b) Missed axle Recognized as 113

FIGURE 12 Computer vision error.

mistake. Although this seems highly unlikely, it happens for one of the following two reasons:

1. B-WIM heuristics coincidentally infers a "ghost axle" at the same position where an actual axle is raised, and the CNN correctly recognizes it. Both systems therefore agree on the axle configuration, although this configuration is incorrectly inferred from strain signals. This is a methodological problem and is independent of CNN performance.
2. Both systems make the same axle grouping mistake when the distance between two adjacent axles is close to the grouping threshold. This problem is specific to CNNs, since different models may learn different axle grouping criteria.

7 | DISCUSSION

7.1 | Key findings

One of the most interesting findings of this research is that popular architecture-based CNNs can learn to recognize axle configurations from low-quality HGV photos with up to 98% accuracy. Given the less-than-ideal angle of our traffic camera—where even our experts were sometimes unsure of the correct axle count and had to analyze the strain signals—the results are quite impressive. This suggests that the proposed CV-based classifier could improve existing strain-based heuristics. However, it is important to note that the precision achieved by either method alone is lower than that of the proposed extended system.

Another notable result is the high precision of the proposed system, which is nearly 100%. This high precision is not entirely unexpected, given that for an error to occur, both axle configuration classifiers must make the same mistake. Since the B-WIM heuristics achieve approximately 96% accuracy and the CNN classifier can achieve about 98% accuracy, we might expect one such mistake in about 1000 cases with the proposed method. This was indeed observed when using the VGG16-based classifier

14678667, 2025, 30, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/mice.70144 by University Of Primorska Up Fvz, Wiley Online Library on [01/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

on the validation set. However, the challenge of learning a fine-tuned threshold for grouping axles under the given camera angle led to a few more mistakes with other CNN architectures. The severity of these mistakes is debatable, as they are likely not as detrimental to the GVW computation as missing or adding an axle. A possible approach to this issue could be fuzzy labeling, where the expert assigns multiple labels to a single instance with a probability score.

7.2 | Limitations of the available data set

We tested our classifier only with photos taken in favorable weather conditions and during daylight. In other words, our experts labeled only those instances where axle configurations were recognizable by the naked eye and could therefore be labeled correctly. As a result, much of the traffic recorded at night, in thick fog, or during heavy rain was excluded from the experiment. While it could be argued that high-quality training samples should be selected to train a model, testing should ideally cover a broader range of scenarios. However, since dark and heavily blurred photos could not be labeled with complete certainty, they could not be used to evaluate the accuracy of the trained classifiers. A possible alternative approach would be to apply photo filters to simulate low-visibility conditions on the labeled clear photos.

In a live scenario, we could either turn off the proposed CV-based discriminator at night and revert to baseline performance, or allow the discriminator to eliminate more vehicles during nighttime, which should, in principle, lower the recall rate but have minimal impact on the system's precision. Another option is to install night vision or thermal cameras. However, in this research, we use existing data acquired with current B-WIM installations.

7.3 | Advantages and limitations of the method

A key limitation of our approach to axle configuration recognition from photographs is its reliance on a predetermined set of axle configuration classes. An axle configuration not included in one of the training classes cannot be verified and is flagged as unreliable a priori. However, such configurations are very rare and are observed only a few times per year, therefore, their exclusion from GVW estimation represents a negligible loss.

On the other hand, recognizing axle configurations as a whole, rather than identifying individual axles and grouping them afterward, offers two important advantages: (1) It eliminates the need for error-prone manual fine-tuning of axle grouping parameters for a specific site and (2) it resolves the issue of car carriers, where the axles of carried cars must be ignored. The first point is supported by our results, in which a trained CNN alone exceeded the accuracy of the existing B-WIM heuristics—manually fine-tuned by engineers for several years—by a few percentage points. However, we recommend training the models specifically for each chosen B-WIM site, as different camera angles, distances, and other visual factors may affect CA.

The problem of recognizing RAs remains a subject for future work. The discrepancy between CV and sensor data is a strong indicator of an RA, especially when considering the typical positions of RAs within axle groups and integrating this knowledge into the system. This should make it possible to evaluate the probability of an RA. However, errors are still possible, and including such cases in the set of reliable GVW measurements would cause a certain drop in precision.

Lastly, there is the question of generalization versus memorization of the trained CNN models. Achieving similar results on a separate validation set compared to cross-validation on the training set may not guarantee generalization, since only a limited number of truck types exist, and many may pass the same B-WIM site daily. Therefore, excluding specific days from training may not prevent the same trucks from appearing in the training data, just recorded on different days. However, memorization may not be a significant practical issue. Even if the model learns to identify axle configurations by memorizing all existing types and brands of vehicles, this does not affect the practical usefulness of the solution, except for the possibility that the models should be retrained every few years as new types of vehicles appear on the roads.

8 | CONCLUSION

This paper presents a low-cost CV extension for commercially available B-WIM systems that can improve the reliability of estimated GVWs from the existing 96.70% to up to 99.89%. This reliability is quantified by the classification precision achieved by the proposed system, in which unreliable GVW estimates are excluded. In practice, this means that 34 out of 35 incorrectly measured vehicles over a 24-h period were correctly identified as unreliable and excluded from further analysis. The proposed solution can be integrated with existing B-WIM installations without disruption or applied retrospectively to previously collected B-WIM data.

The main limitation of the proposed method is the inability of the CV models to recognize RAs under the typical visual conditions of traffic monitoring cameras. Our compromise is to have the method exclude vehicles with

14678667, 2025, 30, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/mice.70144 by University Of Primorska Up Fvz, Wiley Online Library on [01/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

RAs as unreliable, since these vehicles are usually lighter and their GVW is less critical for monitoring structural deterioration. Future work could address this problem by investigating the use of multimodal learning (Erukude et al., 2025) to combine strain signals with camera images, or Dynamic Ensemble Learning (Alam et al., 2020) to combine several CNN models.

ACKNOWLEDGMENTS

The authors would like to express their sincere gratitude to the Motorway Company of the Republic of Slovenia (DARS d.d.) for their valuable support in this study. Special thanks are also extended to CESTEL d.o.o. for their assistance related to the SiWIM system. The authors would like to recognize the contributions of Neža Gošte and Mirko Kosič for their support in building the ground truth data.

This research was funded by The Slovenian Research and Innovation Agency, grant numbers P5-0433, IO-0035, J5-50155 and J7-50096. This work has also been supported by the research programs CogniCom (0013103) at the University of Primorska and the Building Structures and Materials (P2-0273) at Slovenian National Building and Civil Engineering Institute.

REFERENCES

Abadi, M., Barham, P., Chen, J., Chen, Z., Davis, A., Dean, J., Devin, M., Ghemawat, S., Irving, G., Isard, M., Kudlur, M., Levenberg, J., Monga, R., Moore, S., Murray, D. G., Steiner, B., Tucker, P., Vasudevan, V., Warden, P., Wicke, M., et al. (2016). Tensorflow: A system for large-scale machine learning. In *Proceedings of the 12th USENIX conference on operating systems design and implementation* (OSDI'16, pp. 265–283), USA. USENIX Association. https://dl.acm.org/doi/10.5555/3026877.3026899

Alam, K. M. R., Siddique, N., & Adeli, H. (2020). A dynamic ensemble learning algorithm for neural networks. *Neural Computing and Applications*, *32*(12), 8675–8690. https://doi.org/10.1007/s00521-019-04359-7

Almutairi, A., He, P., Rangarajan, A., & Ranka, S. (2022). Automated truck taxonomy classification using deep convolutional neural networks. *International Journal of Intelligent Transportation Systems Research*, *20*(2), 483–494. https://doi.org/10.1007/s13177-022-00306-4

Chang, L., Tang, Q., Xin, J., Jiang, Y., Zhang, H., Li, Z., Zhou, Y., & Zhou, J. (2025). Low-complexity real-time detection Transformer for identifying bridge vehicle loads. *Computer-Aided Civil and Infrastructure Engineering*, *40*(26), 4485–4506. https://doi.org/10.1111/mice.70061

Deng, J., Dong, W., Socher, R., Li, L.-J., Kai, L., & Li, F.-F. (2009). ImageNet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition* (pp. 248–255), Miami, FL. IEEE. https://doi.org/10.1109/CVPR.2009.5206848

Erukude, S. T., Veluru, S. R., & Marella, V. C. (2025). Multimodal deep learning: A survey of models, fusion strategies, applications, and research challenges. *International Journal of Computer Applications*, *187*(19), 1–7. https://doi.org/10.5120/ijca2025925264

Feng, J., Gao, K., Zhang, H., Zhao, W., Wu, G., & Zhu, Z. (2024). Non-contact vehicle weight identification method based on explainable machine learning models and computer vision. *Journal of Civil Structural Health Monitoring*, *14*(4), 843–860. https://doi.org/10.1007/s13349-023-00757-7

Feng, M. Q., & Leung, R. Y. (2021). Application of computer vision for estimation of moving vehicle weight. *IEEE Sensors Journal*, *21*(10), 11588–11597. https://doi.org/10.1109/JSEN.2020.3038186

Feng, M. Q., Leung, R. Y., & Eckersley, C. M. (2020). Non-contact vehicle weigh-in-motion using computer vision. *Measurement*, *153*, 107415. https://doi.org/10.1016/j.measurement.2019.107415

Gao, K., Zhang, H., & Wu, G. (2024). A multispectral vision-based machine learning framework for non-contact vehicle weigh-in-motion. *Measurement*, *226*, 114162. https://doi.org/10.1016/j.measurement.2024.114162

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Identity mappings in deep residual networks. In Leibe, B., Matas, J., Sebe, N., & Welling, M., (Eds.), *Computer vision—ECCV 2016* (Vol. 9908, pp. 630–645). Springer International Publishing. https://doi.org/10.1007/978-3-319-46493-0_38

Howard, A., Sandler, M., Chen, B., Wang, W., Chen, L.-C., Tan, M., Chu, G., Vasudevan, V., Zhu, Y., Pang, R., Adam, H., & Le, Q. (2019). Searching for MobileNetV3. In *2019 IEEE/CVF international conference on computer vision (ICCV)* (pp. 1314–1324), Seoul, Korea (South). IEEE. https://doi.org/10.1109/ICCV.2019.00140

Huang, G., Liu, Z., Van Der Maaten, L., & Weinberger, K. Q. (2017). Densely connected convolutional networks. In *2017 ieee conference on computer vision and pattern recognition (CVPR)* (pp. 2261–2269), Honolulu, HI. IEEE. https://doi.org/10.1109/CVPR.2017.243

Jacob, B., & Feypell-de La Beaumelle, V. (2010). Improving truck safety: Potential of weigh-in-motion technology. *IATSS Research*, *34*(1), 9–15. https://doi.org/10.1016/j.iatssr.2010.06.003

Kingma, D. P., & Ba, J. (2015). Adam: A method for stochastic optimization. In *Proceedings of the 3rd International Conference on Learning Representations (ICLR 2015)*. https://arxiv.org/abs/1412.6980

Kirushnath, S., & Kabaso, B. (2018). Weigh-in-motion using machine learning and telematics. In *2018 2nd international conference on telematics and future generation networks (TAFGEN)* (pp. 115–120), Kuching. IEEE. https://doi.org/10.1109/TAFGEN.2018.8580463

Kong, X., Zhang, J., Wang, T., Deng, L., & Cai, C. (2022). Non-contact vehicle weighing method based on tire-road contact model and computer vision techniques. *Mechanical Systems and Signal Processing*, *174*, 109093. https://doi.org/10.1016/j.ymssp.2022.109093

Meiring, G., & Myburgh, H. (2015). A review of intelligent driving style analysis systems and related artificial intelligence algorithms. *Sensors*, *15*(12), 30653–30682. https://doi.org/10.3390/s151229822

Moses, F. (1979). Weigh-in-motion system using instrumented bridges. *Transportation Engineering Journal of ASCE*, *105*(3), 233–249. https://doi.org/10.1061/TPEJAN.0000783

Pais, J. C., Amorim, S. I. R., & Minhoto, M. J. C. (2013). Impact of traffic overload on road pavement performance. *Journal of Transportation Engineering*, *139*(9), 873–879. https://doi.org/10.1061/(ASCE)TE.1943-5436.0000571

14678667, 2025, 30, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/mice.70144 by University Of Primorska Up Fvz, Wiley Online Library on [01/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Rafiei, M. H., & Adeli, H. (2018). Novel machine-learning model for estimating construction costs considering economic variables and indexes. *Journal of Construction Engineering and Management*, *144*(12), 04018106. https://doi.org/10.1061/(ASCE)CO.1943-7862.0001570,

Rafiei, M. H., Khushefati, W. H., Demirboga, R., & Adeli, H. (2017). Supervised deep restricted Boltzmann machine for estimation of concrete. *ACI Materials Journal*, *114*(2). https://doi.org/10.14359/51689560

Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2016). You Only Look Once: Unified, real-time object detection. In *2016 IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 779–788), Las Vegas, NV, USA. IEEE. https://doi.org/10.1109/CVPR.2016.91

Rys, D., Judycki, J., & Jaskula, P. (2016). Analysis of effect of overloaded vehicles on fatigue life of flexible pavements based on weigh in motion (WIM) data. *International Journal of Pavement Engineering*, *17*(8), 716–726. https://doi.org/10.1080/10298436.2015.1019493

Shah, R., Sharma, Y., Mathew, B., Kateshiya, V., & Parmar, J. (2016). Review paper on overloading effect. *International Journal of Advanced Scientific Research and Management*, *1*(4), 131–134.

Simonyan, K., & Zisserman, A. (2015). Very deep convolutional networks for large-scale image recognition. In *3rd international conference on learning representations (ICLR 2015)* (pp. 1–14). San Diego, CA, USA, May 7–9, 2015, Conference Track Proceedings. http://arxiv.org/abs/1409.1556

Slootmans, F. (2023). *Facts and figures—Buses / coaches / heavy goods vehicles—2023*. Technical report, European Commission, Directorate General for Transport, Brussels, Belgium.

Sujon, M., & Dai, F. (2021). Application of weigh-in-motion technologies for pavement and bridge response monitoring: State-of-the-art review. *Automation in Construction*, *130*, 103844. https://doi.org/10.1016/j.autcon.2021.103844

Wang, Z., Zhu, J., & Ma, T. (2024). Deep learning–based detection of vehicle axle type with images collected via UAV. *Journal of Transportation Engineering, Part B: Pavements*, *150*(3), 04024032. https://doi.org/10.1061/JPEODX.PVENG-1524

Xu, Z., Wei, B., & Zhang, J. (2023). Reproduction of spatial–temporal distribution of traffic loads on freeway bridges via fusion of camera video and ETC data. *Structures*, *53*, 1476–1488. https://doi.org/10.1016/j.istruc.2023.05.023

Yan, W., Ren, H., Luo, X., & Li, S. (2025). Hybrid-data-driven bridge weigh-in-motion technology using a two-level sequential artificial neural network. *Computer-Aided Civil and Infrastructure Engineering*, *40*(20), 2992–3012. https://doi.org/10.1111/mice.13415

Yang, J., Bao, Y., Sun, Z., & Meng, X. (2024). Computer vision-based real-time identification of vehicle loads for structural health monitoring of bridges. *Sustainability*, *16*(3), 1081. https://doi.org/10.3390/su16031081

Zhang, H., Zhu, J., Zhou, Y., & Shen, Z. (2024). Non-contact weigh-in-motion approach with an improved multi-region of interest method. *Mechanical Systems and Signal Processing*, *212*, 111323. https://doi.org/10.1016/j.ymssp.2024.111323

Zhang, J., Kong, X., OBrien, E. J., Peng, J., & Deng, L. (2023). Noncontact measurement of tire deformation based on computer vision and Tire-Net semantic segmentation. *Measurement*, *217*, 113034. https://doi.org/10.1016/j.measurement.2023.113034

Zhang, J., OBrien, E. J., Kong, X., & Deng, L. (2024). Factors affecting the accuracy of a computer vision-based vehicle weight measurement system. *Measurement*, *224*, 113840. https://doi.org/10.1016/j.measurement.2023.113840

Žnidarič, A., & Kalin, J. (2020). Using bridge weigh-in-motion systems to monitor single-span bridge influence lines. *Journal of Civil Structural Health Monitoring*, *10*(5), 743–756. https://doi.org/10.1007/s13349-020-00407-2

Žnidarič, A., Kalin, J., & Kreslin, M. (2018). Improved accuracy and robustness of bridge weigh-in-motion systems. *Structure and Infrastructure Engineering*, *14*(4), 412–424. https://doi.org/10.1080/15732479.2017.1406958

How to cite this article: Šoberl, D., Kalin, J., Anžlin, A., Kreslin, M., Čopič Pucihar, K., Kljun, M., Hekič, D., & Žnidarič, A. (2025). Enhanced precision in axle configuration inference for bridge weigh-in-motion systems using computer vision and deep learning. *Computer-Aided Civil and Infrastructure Engineering*, *40*, 6201–6216. https://doi.org/10.1111/mice.70144

14678667, 2025, 30, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/mice.70144 by University Of Primorska Up Fvz, Wiley Online Library on [01/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License